GEOMETRY · TRIANGLE CONGRUENCE

Triangle Congruence Builder

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Triangle Congruence Builder** on Mr. K's Classroom: Geometry → Triangle Congruence → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. Do you think two triangles with the same three angles must be congruent?

   No. They could be different sizes.

2 Explore use the tool and record what you see

Build each set of parts

Choose the shortcut, set the values and count the triangles.

Given	Values	How many triangles?	Proves congruence?
SSS	6, 5, 7	**1**	**yes**
SSS	2, 2, 7	**0**	**— (no triangle)**
SAS	AC = 5, ∠A = 50°, AB = 7	**1**	**yes**
ASA	∠A = 50°, AB = 7, ∠B = 60°	**1**	**yes**
AAS	∠A = 50°, ∠B = 60°, BC = 6	**1**	**yes**
SSA	AC = 7, BC = 5, ∠A = 40°	**2**	**no**
SSA	AC = 7, BC = 8, ∠A = 40°	**1**	**not always (SSA can fail)**
AAA	∠A = 50°, ∠B = 60°	**infinitely many**	**no**

3 Write the rule fill in each blank from what you observed

1. SSS, SAS, ASA and AAS each build exactly **one** triangle, so they prove congruence.
2. SSA can build **two** different triangles, so it is not a congruence shortcut.
3. AAA fixes a triangle's **shape** but not its **size**.
4. In SAS, the angle must be **between (included by)** the two sides.
5. Three lengths make a triangle only if the two shorter ones add to **more than** the longest.

4 Check no tool: show your work

1. Two sides and the included angle are congruent. Which shortcut?

   SAS

2. ∠A ≅ ∠D, ∠B ≅ ∠E and BC ≅ EF. Which shortcut?

   AAS

3. AB ≅ DE, BC ≅ EF and AC ≅ DF. Which shortcut?

   SSS

4. AB ≅ DE, BC ≅ EF and ∠A ≅ ∠D. Is that enough?

   No: that is SSA

5 Reflect

Explain why SSA can make two different triangles.

The side across from the given angle can swing to meet the base in two places, so the same parts fit two different triangles.